

don't bother about this, and the split seems to be between AI workers—who build, say, intelligent controllers, and take Strong AI for granted—and cognitive scientists, who study what understanding and consciousness are.

John Searle sparked off a debate in the 1980s, which openly challenged the likes of Minsky, with his famous Chinese-Room thought experiment (see box 'Searle's Chinese Room'). It's a very simple argument, but it led to a fabulous number of debates. So many that even today, people devote large amounts of space in Masters and Doctoral theses to discuss the Chinese Room and its counter-arguments.

Machine understanding?

So, do machines understand what they are doing? Can they? Does it make sense to say that a machine is thinking, feeling, and understanding? Strong Alers says that they can; but proponents of other AI viewpoints, as well as many in the general public, say no. One thing that propels people into not believing in strong AI is that people don't like being compared to machines. Some feel indignant if it is suggested that they might be nothing more than something that can be built in a laboratory. That 'nothing more' is at the heart of the problem, philosophically speaking: we have not yet found what it means to be human.

Whereas the previous holy grail of AI, the wonder-robot, has been replaced by the real mystery of consciousness and awareness. You can get a flavour for the kind of debates that rage on at sites such as <http://KurzweilAI.net>, <http://edge.org>, <http://AI.about.com>, and others. Hence, although the state of the art of AI reach-

Searle's Chinese Room

Searle's Chinese Room is a thought experiment, i.e., an argument that one ponders, and draws conclusions from. It involves the idea that even if a robot does things that seem intelligent, such as answering questions about a story it has been told, it does not really understand anything at all.

Imagine a Chinese story-understanding system, where a man who knows no Chinese at all is sitting inside a closed room. This man corresponds to the robot's understanding. He communicates with the world by reading slips of paper, with Chinese on them, thrown in to him in his

room—which correspond to the questions about a story input to the robot. He throws out slips of paper—the answers to those questions—in Chinese, according to rules that have been assigned to him. These rules correspond to the rules according to which a robot might operate. So, when the system is working, the man receives slips of paper, looks up his rule book, and throws out appropriate paper slips. Hence, it appears to a Chinese person, as though the system is doing something intelligent; but the man inside understands nothing of the story at all!

es out to such stupendous achievements as intelligent controllers guiding satellites near Mars, all by themselves, a lot of the focus today is still in understanding how our brains work. Thousands of people are studying how cognition takes place. Research in this area does not actually produce gizmos; but it is a science that will last us a long while, and it is one of the most exciting fields in AI today.

Will robots inherit the earth?

As Minsky said, the true majesty of general intelligence still awaits our attack. Explore some of the material out there, and come to your own conclusions: Is robot intelligence possible? But the point is that we may not need to build one super-robot for Minsky's Dream to come alive; it can be done in stages. We already have people who augment their bodies at

the tissue and sensor level. If tissues can be replaced, what about major organs? And when it comes to the brain, we already know that certain areas of the brain control certain aspects of our senses and perceptions; electrodes placed at specific locations can cause specific sensations. So can we not have an array of electrodes, so that we can feel exactly the way we want to feel?

We also know that our thoughts are governed by certain nerve impulses. If the specifics were mapped out, may we not be able to think in a particular way, whenever we want, wherever we want? What about attaching various thought-modules to ourselves, to be used as and when required? And hence, in the near future, can we expect our race to be composed of entities that are part human, part robot? And after that, can we anticipate pure robots, without any organic tissue? Where is the dividing line?

Equally compelling are the questions and arguments of nay-sayers: scientists and thinkers who believe that we will never be able to adequately explain how our brains work; that mechanical devices in our brains may make them faster or change our mood, but will never be able to make a fundamental difference; that no machine will ever be able to understand things the way we do; and so on.

These are profound issues, these are interesting times, and what was science fiction, may soon be reality. Of course, we've heard that phrase several times before: "what was fiction may become reality". But there's a difference here. We're talking about ourselves.

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